United International University (UIU)
Dept. of Computer Science & Engineering (CSE)
Final-Term Exam: Trimester: Spring 2026

Course Code: EEE 2123; **Course Title:** Electronics
Total Marks: 40; **Duration:** 2 **Hours**

Any examinee found adopting unfair means would be expelled from the trimester/ program as per UIU disciplinary rules.

There are six (6) questions in this paper. Answer all of them.

1. In the circuit of ***Fig 1***, assume a constant **200 µA** current flows through ***R***. If both the MOSFETs have $\mu_n C_{ox}$ = **100 µA/V²**, ***W/L*** = **20** then determine the values of V_1 , V_2 and V_3 . *(Hint: 200µA Current will be equally divided between the 2 mosfets)* **[7]**

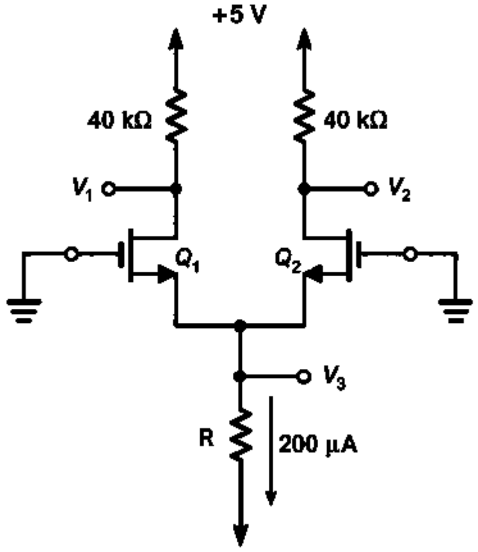

Fig 1: Figure for question 1

2. Analyze the circuit in **Fig 2** to determine voltages at all nodes and the currents through all branches. Let, V_{tn} = **1V** , k_n' **W/L = 1 mA/V²** **[8]**

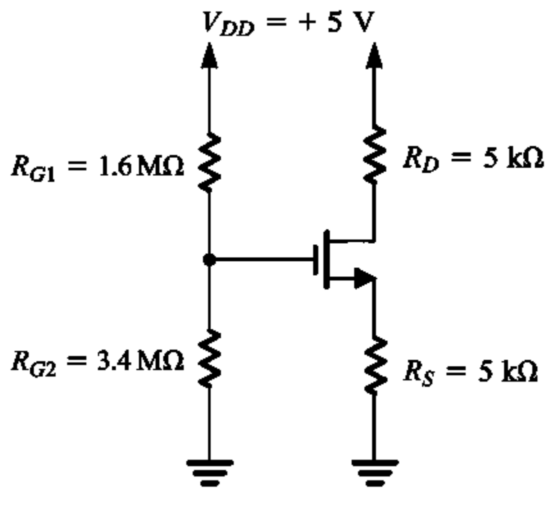

Fig 2: Figure for question 2

3. From the circuit of **Fig 3**, Find all node voltages and branch currents in the following circuit, $\boldsymbol{\beta = 120}$ **[6]**

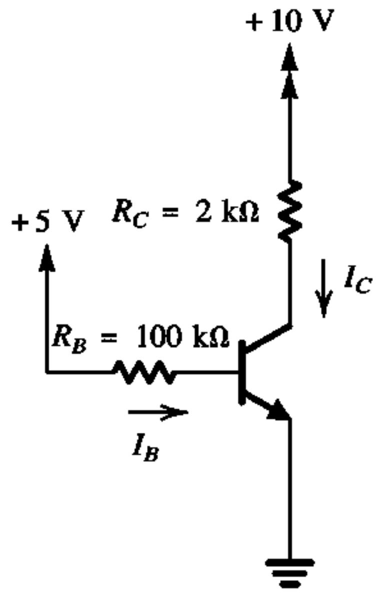

Fig 3: Figure for question 3

4. The BJT in **Fig 4** is operating at the ***edge of saturation*** (V_{CE} = 0.3 V, β = 100) . If R_{B1} = 70 kΩ, determine the value of R_{B2}, all node voltages and branch currents. **[7]**

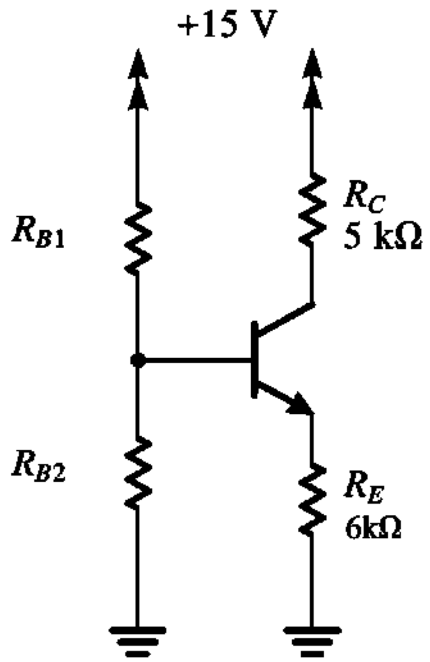

Fig 4: Figure for question 4

5. a) A battery-powered wearable device uses CMOS technology for its digital circuits. Explain why CMOS is a suitable choice for this application in terms of power efficiency. **[2]**

   b) Implement the following boolean expression using CMOS technology. You **cannot use more than** 4 NMOS and 4 PMOS. **[4]**

$$F = \overline{(A+B)(A+C\overline{B}+CB\overline{D}+CBD)}$$

6. a) In the Op-Amp circuit of **Fig 6**, V_1 and V_2 are inputs and V_o is the output. The circuit has the output expression given below. Find the values of R and C in the circuit. **[3]**

$$V_o = 2\frac{dV_2}{dt} - 6V_1$$

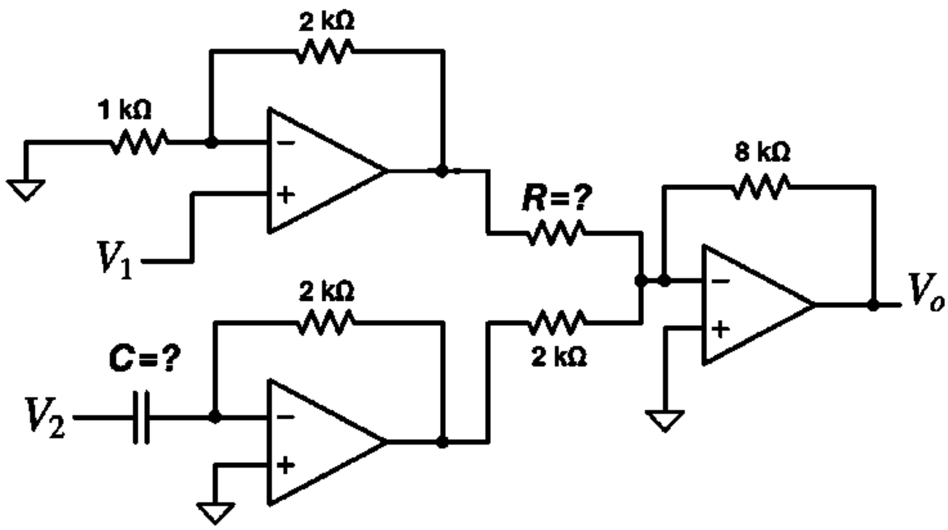

Fig 6: Figure for question 6(a).

   b) Implement the following expression using op-amps. **[3]**

$$V_o = 4V_1 + 7V_2$$